Lab 4

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Task 1: Build first HTML file with Fundamental HTML Document and HTML tags

Created the folder "CS471_lab" in the "libraryproject" folder. Then added the four html files.

`index.html`:

libraryproject > CS471_lab > index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>My First Index HTML Page</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <meta charset="UTF-8">
</head>
<body>
    <div>
        <a href="index.html">Home</a>
        <a href="list_books.html">List Books</a>
        <a href="aboutus.html">About</a>
    </div>
    <h1>This is heading 1 (Library)</h1>
    <p>This is a paragraph text </p>
    <a href="https://www.w3schools.com/html/default.asp">Learn more about HTML</a>
    <hr>
    <!-- This is a comment in HTML, not render in the bowser -->
    <h2>This is heading 2</h2>
    <p>This is some other text. <br> with newline <br> and another line</p>
    <hr>
    <h2>This is heading 2</h2>
    <p>This is some other text.</p>
</body>
</html>
```

`aboutus.html`:

libraryproject > CS471_lab > aboutus.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>About Us</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <meta charset="UTF-8">
</head>
<body>
    <div>
    </div>
    <a href="index.html">Home</a>
    <a href="list_books.html">List Books</a>
    <a href="aboutus.html">About</a>
    <h1>About us</h1>
    <p>This is a paragraph text This is a paragraph text This is a paragraph text
    This is a paragraph text </p>
    <p>To write to <a href="mailto:email@domain.com">our store</a>, click the link
    and your default email client will open an email message and address it to us. </p>
</body>
</html>
```

`list_books.html`:

libraryproject > CS471_lab > list_books.html > html

```html
<!DOCTYPE html>
<html lang="en">
    <head>
    <title>List of Books</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <meta charset="UTF-8">
</head>
<body>
    <div>
    </div>
    <a href="index.html">Home</a>
    <a href="list_books.html">List Books</a>
    <a href="aboutus.html">About</a>
    <h1>List of Books</h1>
    <a href="one_book.html"><h2>Internet & World Wide Web How to Program</h2></a>
    <a href="one_book.html"><img src="book1.png" alt="Book1 img desc" width="200"
    height="200"></a>
    <hr>
    <h2>2) C++ How to Program, Late Objects Version</h2>
    <img src="book2.png" alt="Book2 img desc" width="200" height="200">
    <hr>
    <h2>3) Images in Another Folder</h2>
    <img src="book3.png" alt="Book3 img desc" width="200" height="200">
    <hr>
</body>
</html>
```

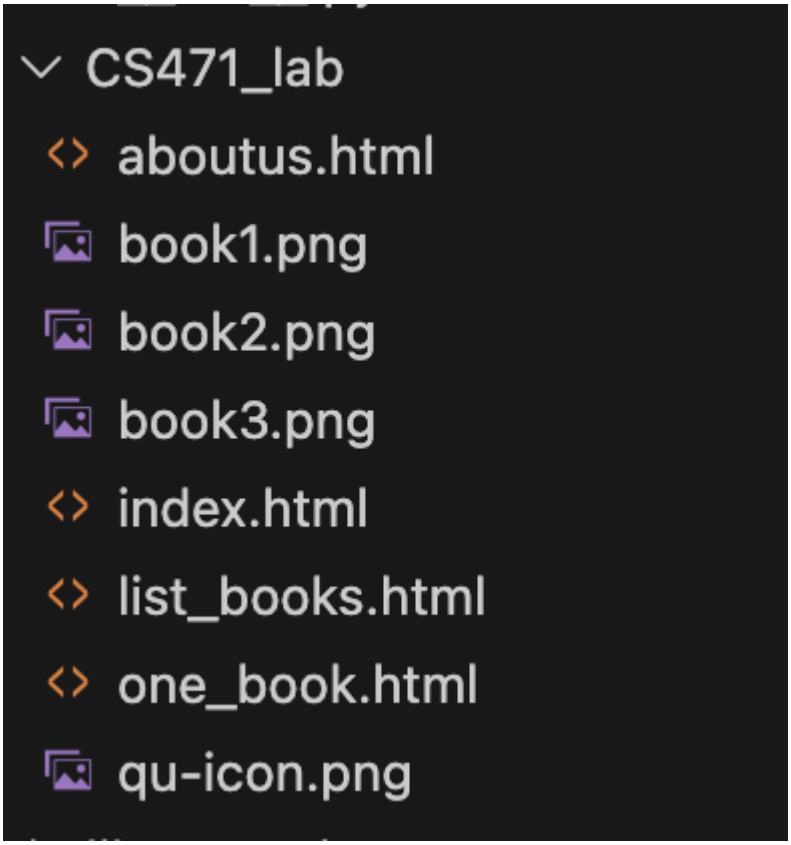
`one_book.html`:

libraryproject > CS471_lab > one_book.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
<title>Show one Book</title>
<meta name="keywords" content="Books, recent, Learning, library">
<meta name="description" content="Books website">
<meta name="author" content="author name">
<link rel="icon" href="qu-icon.png">
<meta charset="UTF-8">
</head>
<body>
<a href="list_books.html">Go Back</a>
<h1>Details of Book1</h1>
<h2>Internet & World Wide Web How to Program</h2>
<h2>Author: author name</h2>
<p><strong> Internet & World Wide Web How to Program, 5/e </strong> provides a
clear, simple, engaging and entertaining introduction to Internet and web
programming. It's appropriate for both introductory and intermediate-level client-
side and server-side programming courses.</p>
<img src="book1.jpg" alt="Book1 img desc" width="200" height="200">
</body>
</html>
```

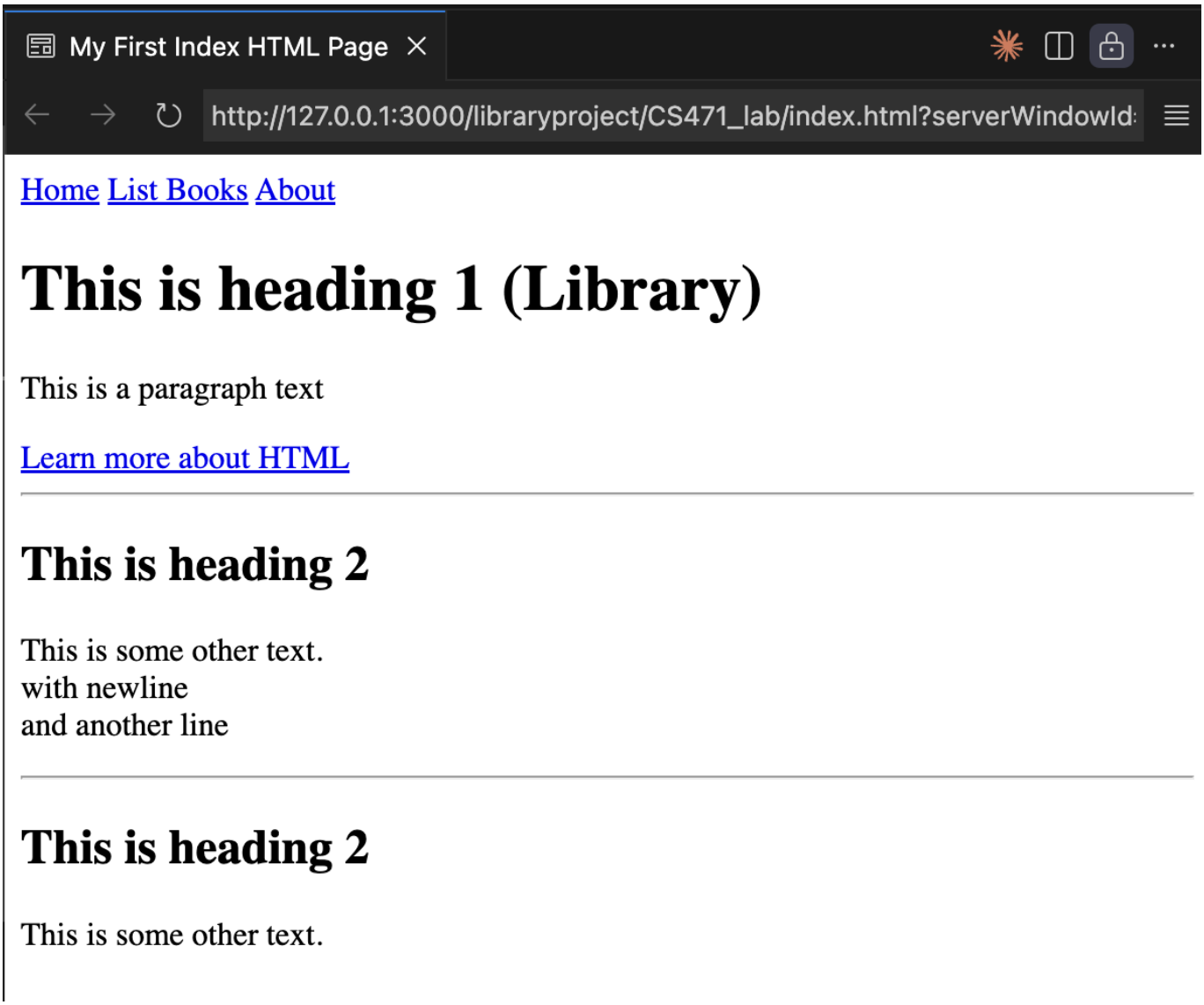
Also added the images `qu-icon.png`, `book1.png`, `book2.png`, and `book3.png` to the same folder.

Folder:



Screenshots:

`index.html`:

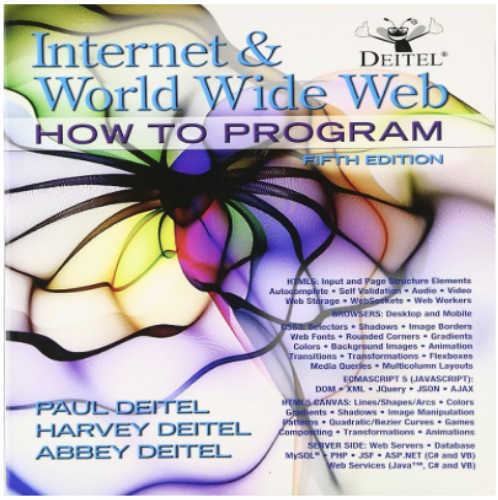


`list_books.html`:

[Home](#) [List Books](#) [About](#)

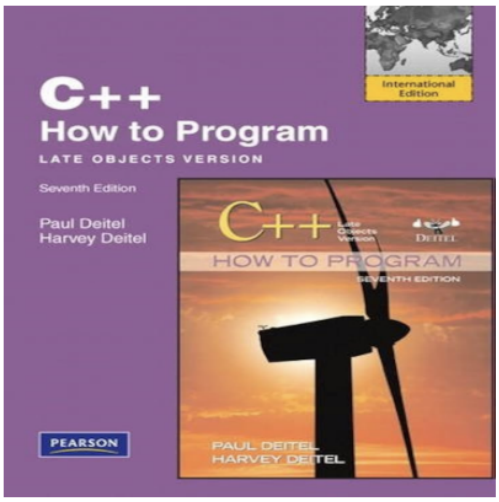
List of Books

[Internet & World Wide Web How to Program](#)



---

2) C++ How to Program, Late Objects Version



---

3) Images in Another Folder

`one_book.html`:

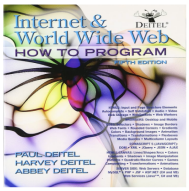
Go Back

Details of Book1

Internet & World Wide Web How to Program

Author: author name

Internet & World Wide Web How to Program, 5/e provides a clear, simple, engaging and entertaining introduction to Internet and web programming. It's appropriate for both introductory and intermediate-level client- side and server-side programming courses.



`aboutus.html`:

Home List Books About

About us

This is a paragraph text This is a paragraph text This is a paragraph text This is a paragraph text

To write to our store, click the link and your default email client will open an email message and address it to us.

Task 2: Style the previous HTML files with CSS in three ways

`index.html`:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>My First Index HTML Page</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <link rel="stylesheet" href="styles.css">
    <meta charset="UTF-8">
</head>
<body>
    <div class="topnav"> <!-- add class attribute -->
        <a class="active" href="index.html">Home</a>
        <a href="list_books.html">List Books</a>
        <a href="aboutus.html">About</a>
    </div>
    <h1 style="color: blue;">This is heading 1 (Library)</h1>
    <p style="color: white;background-color: black;">This is a paragraph
text </p>
    <a href="https://www.w3schools.com/html/default.asp">Learn more about
```

```
HTML</a>
    <hr>
    <h2>This is heading 2</h2>
    <p>This is some other text. <br> with newline <br> and another
line</p>
    <hr>
    <h2>This is heading 2</h2>
    <p>This is some other text.</p>
</body>
</html>
```

list_books.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>List of Books</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <meta charset="UTF-8">
    <style> /* add internal css */
        h1{
        color: green;
        }
        h2{
        color: lightskyblue;
        }
        p {
        font-family: verdana;
        font-size: 20px;
        }
    </style>
    <link rel="stylesheet" href="styles.css"> <!-- include css file -->
</head>
<body>
    <div class="topnav"> <!-- add class attribute -->
        <a href="index.html">Home</a>
        <a class="active" href="list_books.html">List Books</a>
        <a href="aboutus.html">About</a>
    </div>
    <h1>List of Books</h1>
    <a href="one_book.html"><h2>Internet & World Wide Web How to
Program</h2></a>
    <a href="one_book.html"><img src="book1.png" alt="Book1 img desc"
width="200"
    height="200"></a>
    <hr>
    <h2>2) C++ How to Program, Late Objects Version</h2>
    <img src="book2.png" alt="Book2 img desc" width="200" height="200">
    <hr>
```

```
    <h2>3) Images in Another Folder</h2>
    <img src="book3.png" alt="Book3 img desc" width="200" height="200">
    <hr>
</body>
</html>
```

one_book.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Show one Book</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <link rel="stylesheet" href="styles.css">
    <meta charset="UTF-8">
</head>
<body>
    <a href="list_books.html">Go Back</a>
    <h1>Details of Book1</h1>
    <h2>Internet & World Wide Web How to Program</h2>
    <h2>Author: author name</h2>
    <p><strong> Internet & World Wide Web How to Program, 5/e </strong>
provides a
    clear, simple, engaging and entertaining introduction to Internet and
web
    programming. It’s appropriate for both introductory and intermediate-
level client-
    side and server-side programming courses.</p>
    <img src="book1.png" alt="Book1 img desc" width="200" height="200">
</body>
</html>
```

aboutus.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>About Us</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="qu-icon.png">
    <link rel="stylesheet" href="styles.css">
    <meta charset="UTF-8">
</head>
<body>
```

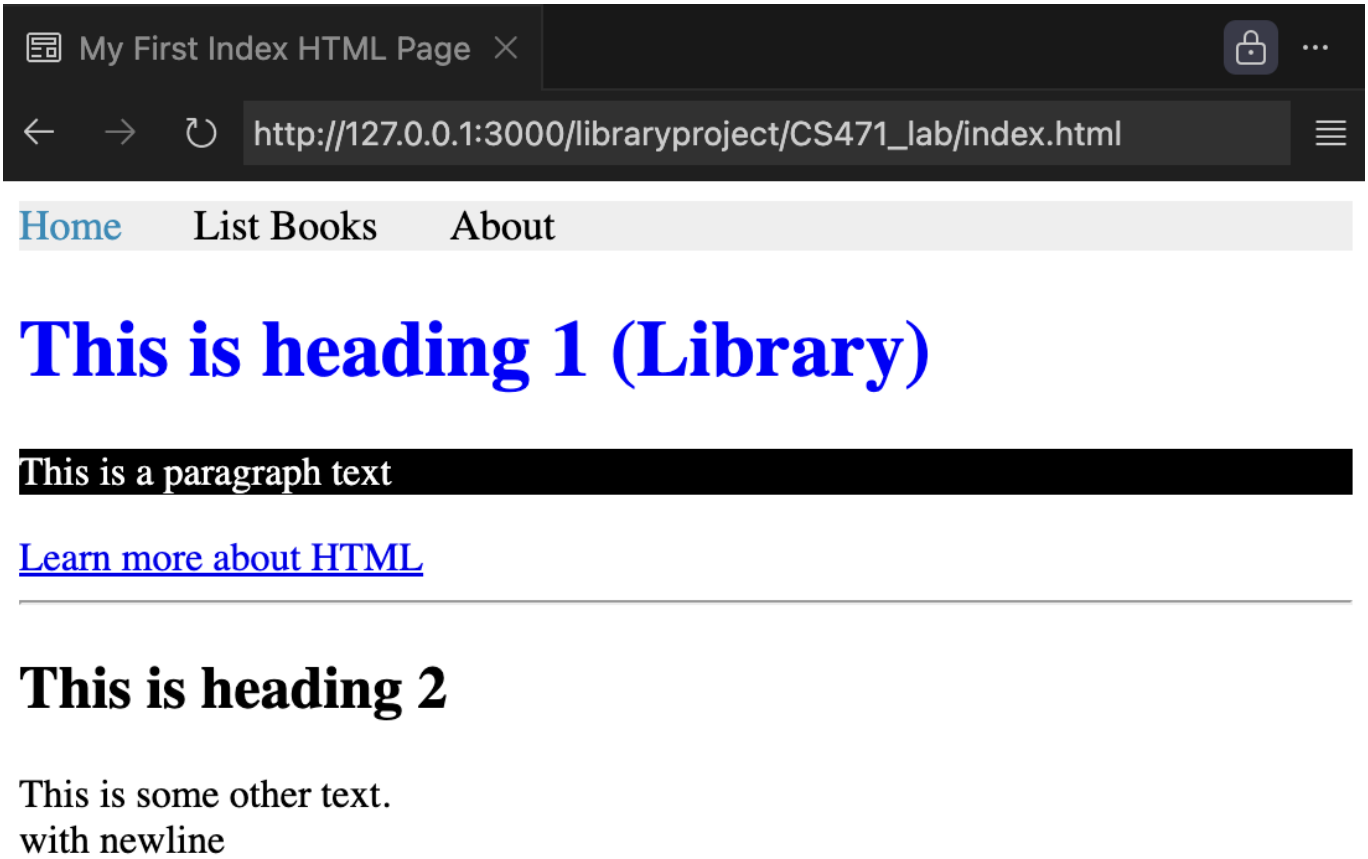
```
    <div class="topnav"> <!-- add class attribute -->
        <a href="index.html">Home</a>
        <a href="list_books.html">List Books</a>
        <a class="active" href="aboutus.html">About</a>
    </div>
    <h1>About us</h1>
    <p>This is a paragraph text This is a paragraph text This is a
paragraph text
    This is a paragraph text </p>
    <p>To write to <a href="mailto:email@domain.com">our store</a>, click
the link
    and your default email client will open an email message and address
it to us. </p>
</body>
</html>
```

styles.css:

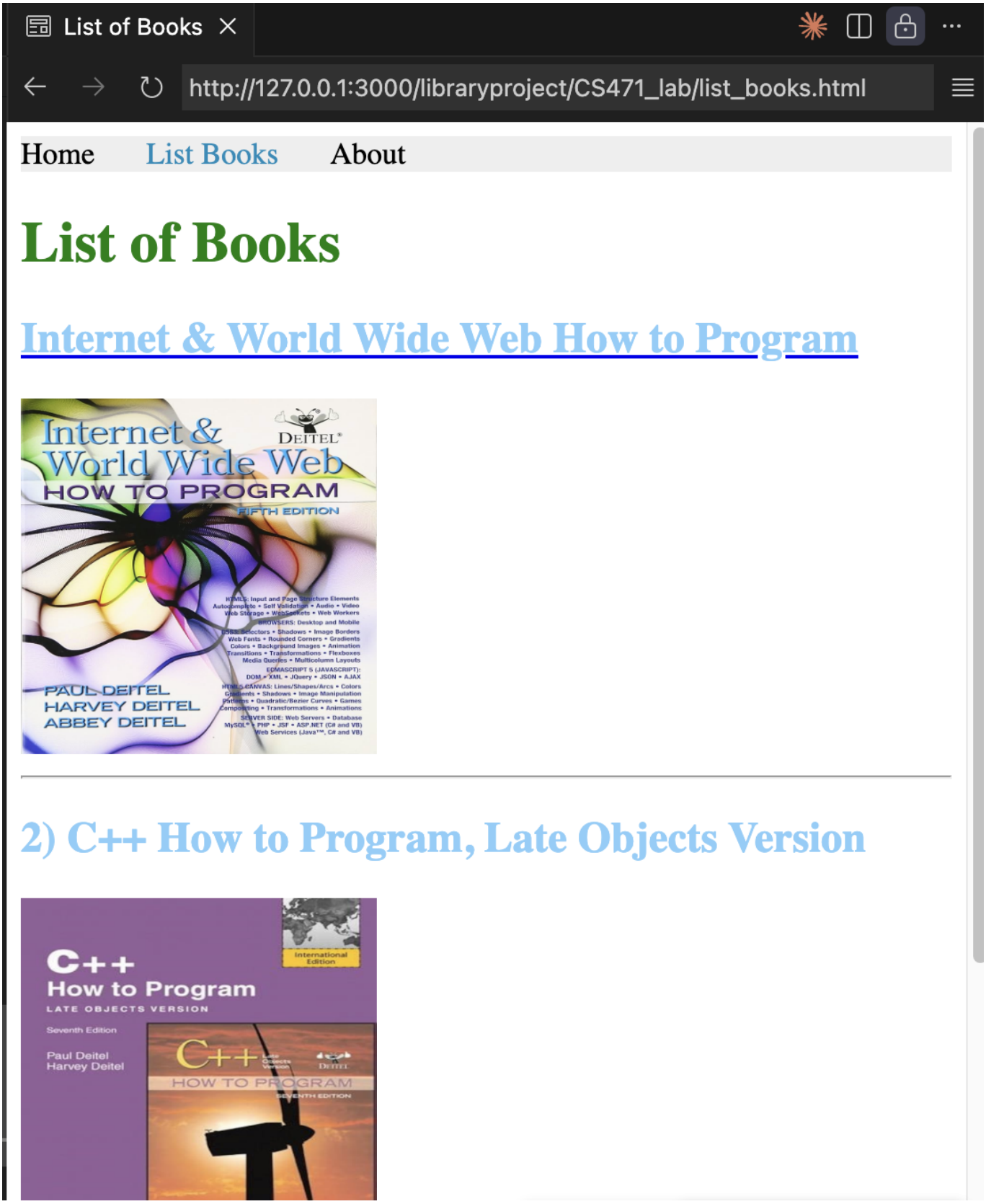
```
.topnav {
    background-color: #eeeeee;
    width: 100%;
}
/* Style the links inside the navigation bar */
.topnav a {
    color: #000000;
    text-align: center;
    margin-right: 25px;
    text-decoration: none;
    font-size: 17px;
}
/* Add a color to the active/current link */
.topnav a.active {
    color: #008CBA;
}
```

Screenshots:

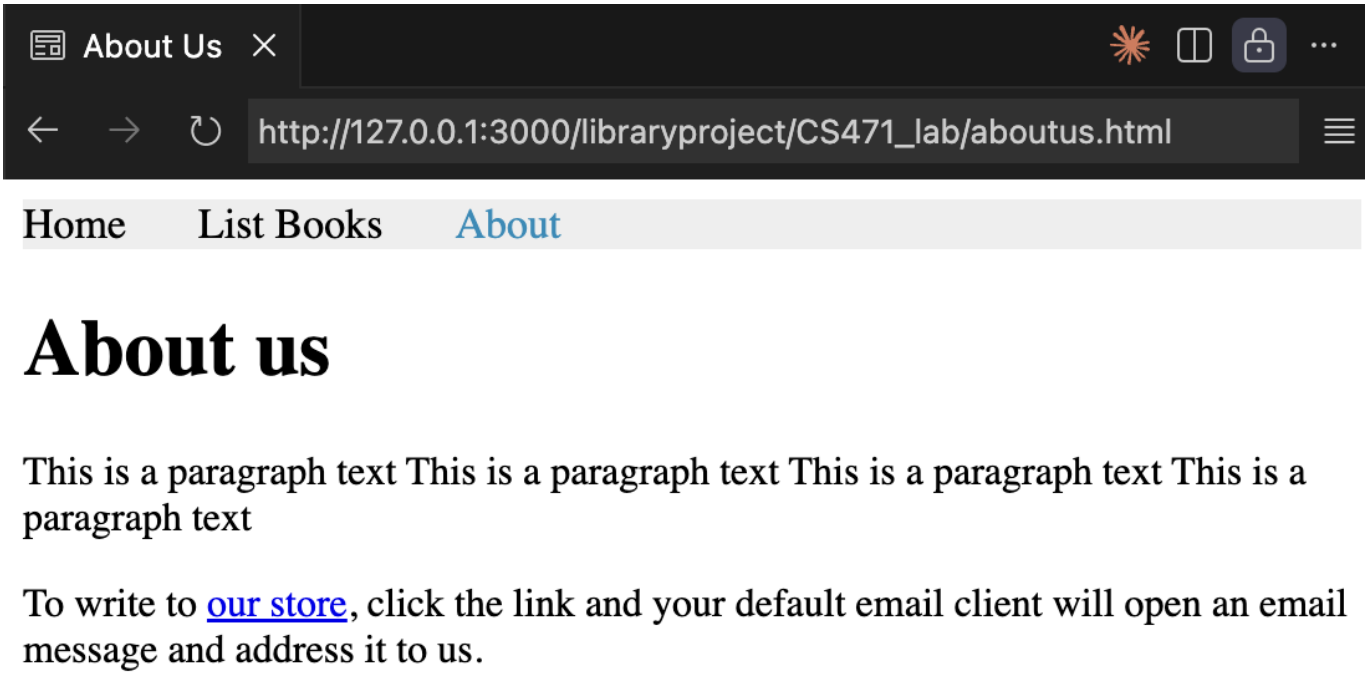
`index.html`:


My First Index HTML Page
http://127.0.0.1:3000/libraryproject/CS471_lab/index.html
Home
List Books
About
This is heading 1 (Library)
This is a paragraph text
Learn more about HTML
This is heading 2
This is some other text.
with newline
and another line
This is heading 2
This is some other text.

`list_books.html`:

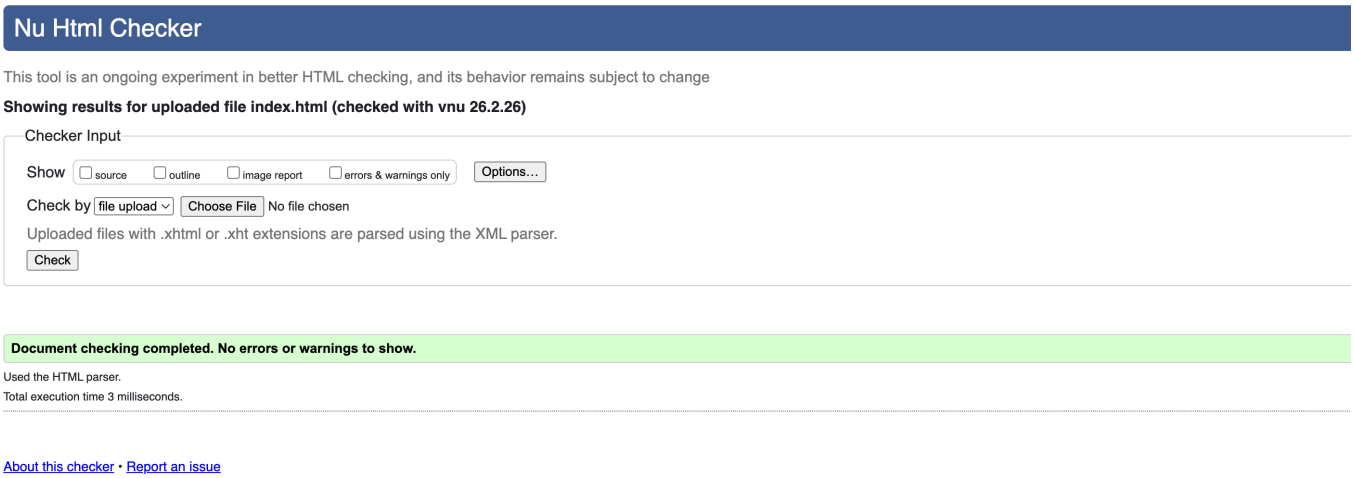


`aboutus.html`:



Task 3: W3C HTML5 Validation Service2 to use proper HTML5 syntax to ensure that browsers process your documents properly

The tool shows no errors in the HTML files.



Task 4: Implementation and management of the previous HTML and static files (e.g. CSS and images) in Django framework

Created `static` folder in the `apps` folder. And updated `settings.py` to include the static files.

```
STATIC_URL = 'static/'
STATICFILES_DIRS = [( os.path.join(BASE_DIR, "apps/static"))]
```

Then moved the `styles.css` and four image files to `apps/static` folder.

Then added the following functions in `apps/bookmodule/views.py`:

```
def index(request):
    return render(request, 'index.html')
def list_books(request):
    return render(request, 'list_books.html')
def viewbook(request, bookId):
    return render(request, 'one_book.html')
def aboutus(request):
    return render(request, 'aboutus.html')
```

And updated the `urls.py` file in the `bookmodule` app to include the paths for the views:

```
urlpatterns = [
    path('', views.index, name= "books.index"),
    path('list_books/', views.list_books, name= "books.list_books"),
    path('<int:bookId>/', views.viewbook, name="books.view_one_book"),
    path('aboutus/', views.aboutus, name="books.aboutus"),
]
```

Then moved HTML files to `apps/templates/bookmodule`, updated URLs (for hyperlink, style, and images) with URL or Static Django template tag.

`index.html`:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <title>My First Index HTML Page</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="{% static 'qu-icon.png' %}">
    <link rel="stylesheet" href="{% static 'styles.css' %}">
    <meta charset="UTF-8">
</head>
<body>
    <div class="topnav"> <!-- add class attribute -->
        <a class="active" href="{% url 'books.index' %}">Home</a>
        <a href="{% url 'books.list_books' %}">List Books</a>
        <a href="{% url 'books.aboutus' %}">About</a>
    </div>
    <h1 style="color: blue;">This is heading 1 (Library)</h1> <!-- add
inline css -->
    <p style="color: white;background-color: black;">This is a paragraph
```

```
text </p>
    <!-- add inline css -->
    <a href="https://www.w3schools.com/html/default.asp">Learn more about
HTML</a>
    <hr>
    <!-- This is a comment in HTML, not render in the bowser -->
    <h2>This is heading 2</h2>
    <p>This is some other text. <br> with newline <br> and another
line</p>
    <hr>
    <h2>This is heading 2</h2>
    <p>This is some other text.</p>
</body>
</html>
```

list_books.html:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <title>List of Books</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="{% static 'qu-icon.png' %}">
    <meta charset="UTF-8">
    <style> /* add internal css */
        h1{
        color: green;
        }
        h2{
        color: lightskyblue;
        }
        p {
        font-family: verdana;
        font-size: 20px;
        }
    </style>
    <link rel="stylesheet" href="{% static 'styles.css' %}"> <!-- include
css file -->
</head>
<body>
    <div class="topnav"> <!-- add class attribute -->
        <a href="{% url 'books.index' %}">Home</a>
        <a class="active" href="{% url 'books.list_books' %}">List
Books</a>
        <a href="{% url 'books.aboutus' %}">About</a>
    </div>
    <h1>List of Books</h1>
    <a href="{% url 'books.view_one_book' 1 %}"><h2>Internet & World Wide
Web How to Program</h2></a>
```

```
    <a href="{% url 'books.view_one_book' 1 %}"><img src="{% static
'book1.png' %}" alt="Book1 img desc" width="200"
    height="200"></a>
    <hr>
    <h2>2) C++ How to Program, Late Objects Version</h2>
    <img src="{% static 'book2.png' %}" alt="Book2 img desc" width="200"
height="200">
    <hr>
    <h2>3) Images in Another Folder</h2>
    <img src="{% static 'book3.png' %}" alt="Book3 img desc" width="200"
height="200">
    <hr>
</body>
</html>
```

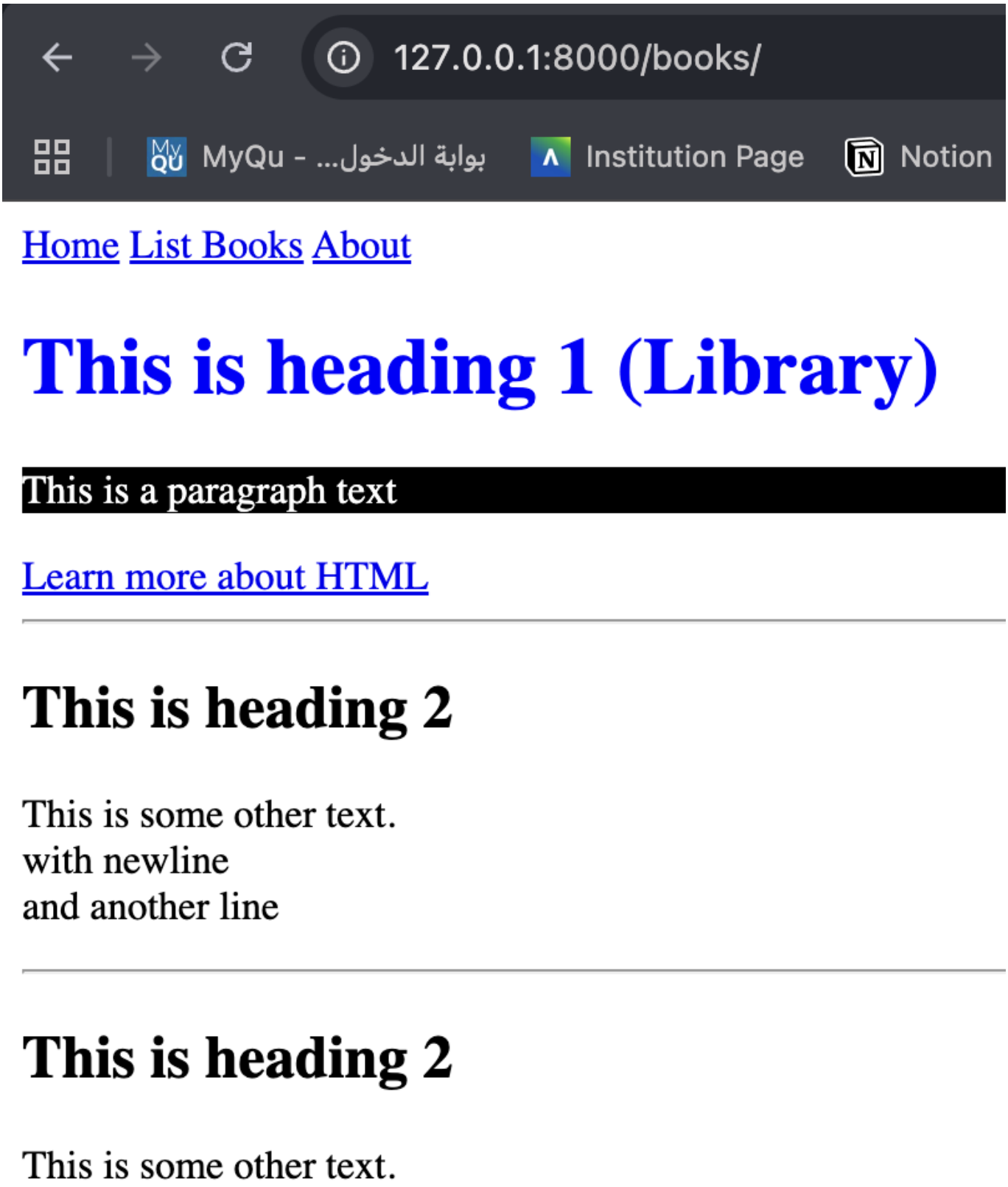
one_book.html:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Show one Book</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="{% static 'qu-icon.png' %}">
    <link rel="stylesheet" href="{% static 'styles.css' %}">
    <meta charset="UTF-8">
</head>
<body>
    <a href="{% url 'books.list_books' %}">Go Back</a>
    <h1>Details of Book1</h1>
    <h2>Internet & World Wide Web How to Program</h2>
    <h2>Author: author name</h2>
    <p><strong> Internet & World Wide Web How to Program, 5/e </strong>
provides a
    clear, simple, engaging and entertaining introduction to Internet and
web
    programming. It's appropriate for both introductory and intermediate-
level client-
    side and server-side programming courses.</p>
    <img src="{% static 'book1.png' %}" alt="Book1 img desc" width="200"
height="200">
</body>
</html>
```

aboutus.html:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <title>About Us</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="{% static 'qu-icon.png' %}">
    <link rel="stylesheet" href="{% static 'styles.css' %}">
    <meta charset="UTF-8">
</head>
<body>
    <div class="topnav"> <!-- add class attribute -->
        <a href="{% url 'books.index' %}">Home</a>
        <a href="{% url 'books.list_books' %}">List Books</a>
        <a class="active" href="{% url 'books.aboutus' %}">About</a>
    </div>
    <h1>About us</h1>
    <p>This is a paragraph text This is a paragraph text This is a
paragraph text
    This is a paragraph text </p>
    <p>To write to <a href="mailto:email@domain.com">our store</a>, click
the link
    and your default email client will open an email message and address
it to us. </p>
</body>
</html>
```

Screenshots after running the server at `http://127.0.0.1:8000/books/`:



Task 5: Additional implementation of Django tags to build one layout instead of repeating the layout for every HTML file

Added the additonal folders in the `templates` folder and created the `base.html` file in the `layouts` folder. Then updated the HTML files to extend the base layout and include the header and footer.

`base.html`:

```
<!DOCTYPE html>
<html>
{% load static %}
<head>
    <title>{% block title %} {% endblock title %}</title>
    <meta name="keywords" content="Books, recent, Learning, library">
    <meta name="description" content="Books website">
    <meta name="author" content="author name">
    <link rel="icon" href="{% static 'qu-icon.png' %}">
    <meta charset="UTF-8">
    <link rel="stylesheet" href="{% static 'styles.css' %}"> <!-- include
css file -->
    {% block stylesheets %}{% endblock stylesheets %}
</head>
<body>
    {% include "includes/header.html" %}
    {% block content %} {% endblock content %}
    {% include "includes/footer.html" %}
</body>
</html>
```

`header.html`:

```
{% with request.resolver_match.url_name as url_name %}
<div class="topnav">
    <a class="{% if url_name == 'books.index' %}active{% endif %}" href="
{% url 'books.index' %}">Home</a>
    <a class="{% if url_name == 'books.list_books' %}active{% endif %}"
href="{% url 'books.list_books' %}">List Books</a>
    <a class="{% if url_name == 'books.aboutus' %}active{% endif %}"
href="{% url 'books.aboutus' %}">About</a>
</div>
{% endwith %}
```

`index.html`:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} My First Index HTML Page {% endblock title %}
{% block content %}
<h1 style="color: blue;">This is heading 1 (Library)</h1> <!-- add inline
css -->
<p style="color: white;background-color: black;">This is a paragraph text
</p> <!--
add inline css -->
<a href="https://www.w3schools.com/html/default.asp">Click Here to learn
more about
HTML & CSS</a>
```

```
<hr>
<!-- This is a comment in HTML, not render in the bowser -->
<h2>This is heading 2</h2>
<p>This is some other text. <br/> with newline <br/> and another line</p>
<hr>
<h2>This is heading 2</h2>
<p>This is some other text.</p>
{% endblock content %}
```

list_books.html:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} List of Books {% endblock title %}
{% block stylesheets %}
<style> /* add internal css */
    h1{
    color: green;
    }
    h2{
    color: lightskyblue;
    }
    p {
    font-family: verdana;
    font-size: 20px;
    }
</style>
{% endblock stylesheets %}
{% block content %}
<h1>List of Books</h1>
<a href="{% url 'books.view_one_book' 1 %}"><h2>Internet & World Wide Web
How to Program</h2></a>
<a href="{% url 'books.view_one_book' 1 %}"><img src="{% static
'book1.png' %}" alt="Book1 img desc" width="200"
height="200"></a>
<hr>
<h2>2) C++ How to Program, Late Objects Version</h2>
<img src="{% static 'book2.png' %}" alt="Book2 img desc" width="200"
height="200">
<hr>
<h2>3) Images in Another Folder</h2>
<img src="{% static 'book3.png' %}" alt="Book3 img desc" width="200"
height="200">
<hr>
{% endblock content %}
```

one_book.html:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} Show one Book {% endblock title %}
{% block content %}
<a href="{% url 'books.list_books' %}">Go Back</a>
<h1>Details of Book1</h1>
<h2>Internet & World Wide Web How to Program</h2>
<h2>Author: author name</h2>
<p><strong> Internet & World Wide Web How to Program, 5/e </strong>
provides a
clear, simple, engaging and entertaining introduction to Internet and web
programming. It's appropriate for both introductory and intermediate-level
client-
side and server-side programming courses.</p>
<img src="{% static 'book1.png' %}" alt="Book1 img desc" width="200"
height="200">
{% endblock content %}
```

`aboutus.html`:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} About Us {% endblock title %}
{% block content %}
<h1>About us</h1>
<p>This is a paragraph text This is a paragraph text This is a paragraph
text
This is a paragraph text </p>
<p>To write to <a href="mailto:email@domain.com">our store</a>, click the
link
and your default email client will open an email message and address it to
us. </p>
{% endblock content %}
```

Screenshots:

Home List Books About

This is heading 1 (Library)

This is a paragraph text

Click Here to learn more about HTML & CSS

This is heading 2

This is some other text.
with newline
and another line

This is heading 2

This is some other text.